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7.4 Trapezoids and Kites

A **trapezoid** has exactly one pair of parallel sides, called the bases. Its **midsegment** (median) connects the midpoints of the two legs, runs parallel to both bases, and has length equal to the average of the two base lengths: $m = \dfrac{b_1 + b_2}{2}$. An **isosceles trapezoid** (congruent legs) has an extra pair of properties: both base angles at each base are congruent to each other, and the diagonals are congruent. A **kite** has two distinct pairs of consecutive congruent sides (not opposite, like a parallelogram). Its diagonals are always perpendicular; the diagonal connecting the vertices between the congruent sides is the axis of symmetry — it bisects the other diagonal *and* bisects the two vertex angles it passes through, while the two angles between the unequal sides are congruent to each other. This worksheet uses all of these relationships together.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Bases 12 and 20. Midsegment?

2. Midsegment 11, base 8. Other base?

3. Base angles $65°$, $65°$. Find the other two base angles.

4. Diagonals 10 and 14. Area?

5. Bases 5 and 17. Midsegment?

6. $\overline{AB} \parallel \overline{DC}$; $m\angle B = 100°$. Find $m\angle C$ (shares leg $\overline{BC}$).

7. Sides $9, 9, 6, 6$. Perimeter?

8. Isosceles trapezoid; legs $4x - 3$ and $2x + 7$. Find x and each leg.

9. Midsegment 18, base 22. Other base?

10. Diagonals of a kite are always…

11. One base angle is $78°$. Find the other angle at the *same* base.

12. Diagonals 9 and 20. Area?

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13. Bases a and b. Write the midsegment formula.

14. Isosceles trapezoid: diagonal $AC = 4x - 6$ and diagonal $BD = 2x + 10$. Find x and the length of each diagonal.

15. Kite $ABCD$ has $m\angle B = m\angle D = 110°$ (the non-vertex angles). If $m\angle A = 3x$ and $m\angle C = x$, find x and both vertex angles.

✓ Answer Key

1. 16
2. 14
3. 115°, 115°
4. 70
5. 11
6. 80°
7. 30
8. $x = 5$; legs = 17
9. 14
10. Perpendicular
11. 78° (base angles at the same base are congruent)
12. 90
13. $m = \dfrac{a + b}{2}$
14. $x = 8$; each diagonal = 26
15. $x = 35°$; $m\angle A = 105°$, $m\angle C = 35°$